

## # AI Governance Complaint Management System - Presentation Content

Use the following content to build your PowerPoint presentation. It covers the MVP definition, how AI is utilized, and a comparison between traditional methods and your innovative solution.

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### ## 1. Minimum Viable Product (MVP) Definition

Our MVP is a unified **\*\*AI-powered Civic Grievance Platform\*\*** designed to connect citizens with their local authorities efficiently.

#### **\*\*Key Features of the MVP:\*\***

- **\*\*Citizen Portal (User):\*\*** A web-based interface for citizens to lodge complaints by providing a text description and uploading image evidence.
- **\*\*Automated AI Processing:\*\*** Immediate, backend AI processing that categorizes the complaint (e.g., Pothole, Water Supply), assigns a priority level, and verifies the authenticity of the uploaded image against the text claim.
- **\*\*Authority Dashboard (Nagar Sevak):\*\*** A localized dashboard for civic officials to view assigned, verified complaints and update their resolution status.
- **\*\*Admin Oversight:\*\*** A high-level view for administrators to track Service Level Agreement (SLA) breaches (Black Zones) and overall grievance resolution metrics via interactive maps and charts.

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### ## 2. Where AI is Used in Our Project

Our system leverages artificial intelligence at the very first touchpoint to automate triage and reduce manual verification efforts.

#### **### A. NLP for Text Classification (Scikit-Learn)**

- **\*\*Category Prediction:\*\*** When a user types a description of their issue, our custom NLP model analyzes the text to automatically categorize the issue into specific departments (e.g., Road/Pothole, Water Supply, Garbage/Sanitation).

- **Priority Assignment:** The same NLP engine assesses the severity of the text description to automatically assign a priority tier (High, Medium, Low), ensuring critical issues are flagged immediately.

### ### B. Computer Vision & VQA for Image Verification (HuggingFace ViLT)

- **Fake/Irrelevant Image Detection:** We use a Visual Question Answering (VQA) model (ViLT-b32). Once the NLP model predicts a category (e.g., "Pothole"), the VQA model dynamically asks the image specific questions (e.g., "Is there a pothole in this image?").
- **Automated Verification:** If the image answers align with the category, the AI marks the complaint as "Verified". If the image is unrelated (e.g., a selfie uploaded instead of a pothole), the AI flags it as "Suspicious", filtering out spam and fake complaints before they reach human officials.

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## ## 3. Old Methods vs. Our Innovative Solution

### ### The Old / Traditional Method

- **Manual Data Entry & Routing:** Citizens fill out lengthy forms, and a human operator reads, interprets, and manually assigns the ticket to the correct department.
- **High Spam & Fake Complaints:** Pranksters or frustrated users upload fake images or irrelevant complaints, wasting the time of field inspectors who travel to the location only to find nothing.
- **Slow Response Times:** Because categorization and verification are manual, critical issues (like open live wires or severe water pipe bursts) sit in the queue waiting to be read.
- **Lack of Transparency:** Citizens submit a complaint into a "black hole" and have no way of tracking if an official is actually working on it.

### ### Our Project's New & Innovative Benefits

- **Zero Human Triage (AI Routing):** AI instantly categorizes and prioritizes the complaint the millisecond it is submitted, routing it to the correct "Nagar Sevak" without human bottleneck.
- **Pre-emptive Spam Filtering:** The AI Vision model acts as a digital inspector. By correlating the image with the text via VQA, it automatically weeds out fake complaints and flags suspicious entries.
- **SLA Tracking & Black Zones:** If an official ignores a high-priority verified complaint, the system maps it as a "Black Zone," forcing accountability and drawing Admin attention to neglected areas.
- **E-Commerce Style Tracking:** Citizens get real-time status updates (Submitted -> AI Verified -> In Progress -> Resolved), building trust between the public and the government.

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## ## 4. Technology Stack (What We Used to Build This)

Here is a breakdown of the core technologies and frameworks used to build the platform:

### ### 📄 Frontend & UI

- **Streamlit**: Used to build the entire web interface (User Portal, Admin Dashboard, and Nagar Sevak dashboards) rapidly using Python.
- **Plotly**: For rendering interactive data visualizations and analytics charts in the Admin dashboard.

### ### ⚙️ Backend & Logic

- **Python 3**: The core programming language running the application logic, routing, and AI integration.
- **MySQL** (`mysql-connector-python`): Relational database used to securely store users, authorities, complaints data, and tracking statuses.

### ### 🧠 Artificial Intelligence & Machine Learning

- **Scikit-Learn (Python)**: Used to train and run the NLP (Natural Language Processing) models (`category\_model.pkl`, `priority\_model.pkl`) that predict text categories and prioritize complaints.
- **Hugging Face Transformers** (`transformers`): Used to load and execute the pre-trained **ViLT** (Vision-and-Language Transformer) model (`vilt-b32-finetuned-vqa`) for Visual Question Answering (VQA) image verification.
- **PyTorch** (`torch`): The deep learning backend framework required to run the Hugging Face vision models efficiently.

### ### 🗺️ Mapping & Geospatial Data

- **Folium & Streamlit-Folium**: Used to generate interactive maps for the Admin dashboard (e.g., plotting complaint locations and SLA "Black Zones").
- **GeoPandas & Pandas**: Used for handling geospatial data points and general data manipulation from the database.